

Q.21 Explain RLC Bridge and describe it's specifications.
(CO4)

Q.22 Write a short note on signature analyzer? (CO5)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the working of a DSO with the help of block diagram in detail. (CO1)

Q.24 Explain the circuit and working of Maxwell's induction bridge and its applications. (CO4)

Q.25 Explain the construction and working of attraction type of Moving Iron Instruments. (CO5)

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Roll No.

5th Sem / Automation & Robotics

Subject : Electronic Instrumentation and Measurements

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Random errors in a measuring system are due to (CO1)

- a) Environmental changes
- b) Use of uncalibrated instrument
- c) Poor cabling practices
- d) Unpredictable effects

Q.2 What is the smallest change in applied input that will indicate a detectable change in output of an instrument? (CO1)

- a) Accuracy
- b) Precision
- c) Sensitivity
- d) Resolution

Q.3 Inductance is measured by which one of the following? (CO4)

- a) Wheat stone bridge b) Schering bridge
c) De Sauty's bridge d) Hay's bridge

Q.4 The wave shape generated by time base circuit of a CRO is (CO3)

- a) Triangular wave b) Square wave
c) Saw tooth wave d) Sine wave

Q.5 Lissajous patterns can be used to measure (CO3)

- a) frequency b) phase difference
c) both a and b d) intensity of light

Q.6 DSO digitises the _____ form of input signal. (CO3)

- a) Digital b) Analog
c) Saw tooth d) None of the above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 The band width of a CRO is measured in _____. (CO3)

Q.8 Q-Factor (Quality Factor) of a coil = _____ (give the formula) (CO3)

Q.9 List any two advantages of Digital multimeter. (CO2)

Q.10 RF stands for _____. (CO5)

Q.11 Sensitivity of a basic meter is measured in _____. (CO4)

Q.12 Name the instrument used to generate sine waves with variable frequency and amplitude _____. (CO4)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 How a Logic Pulser is useful in digital circuits? (CO1)

Q.14 Write four applications of CRO. (CO2)

Q.15 Describe De Sauty's bridge. (CO4)

Q.16 Explain the working principle of Q meter. (CO4)

Q.17 Describe the working of a CRO. (CO2)

Q.18 Elaborate the parts of a CRT with the aid of a diagram. (CO3)

Q.19 How the range of an ammeter is extended? (CO2)

Q.20 What is the use of a Logic Probe? (CO3)